# Lecture 10, Math 441

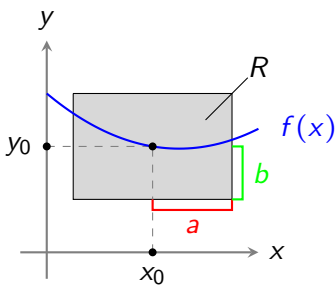
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# Theorem of Existence and Uniqueness for 1st Order IVP

We want to prove the following theorem:

Given the IVP $y' = f(x, y)$, $y(x_0) = y_0$, if $\exists$ a rectangle $R$, $|x - x_0| < a$ and $|y - y_0| < b$, where $f$ and $f_y$ are continuous then $\exists$ an interval $|x - x_0| < h$ where $\exists$ one and only one solution to the IVP.




$h \leq a$

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Without loss of generality we can set $x_0 = 0$ and $y_0 = 0$ so that the IVP is now

$$\begin{cases} y' = f(x, y) \\ y(0) = 0 \end{cases}$$

Also, we can integrate the equation from 0 to $x$ and use $y(0) = 0$ to get

$$y(x) = \int_0^x f(s, y(s)) \, \mathrm{d}s$$

This is the equivalent integral equation.

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# Method of Successive Approximations

Following Picard we construct a sequence of functions.

We can start with $y_0(x) = 0$.

Then we define $y_1(x) = \int_0^x f(s, y_0(s))\,\mathrm{d}s$,

and in general $y_n(x) = \int_0^x f(s, y_{n-1}(s))\,\mathrm{d}s$.

We want to prove that the limit of this sequence exists and it is

the solution to the IVP:

$y_0(x), y_1(x), y_2(x) \cdots y_n(x) \to y(x)$.

First we want to prove that each term of the sequence exists. This is true if the integrand $f$ is continuous in the interval from 0 to $x$.

For $y_0 = 0$ this is true, because $f(s, 0)$ continuous as long as $|s| < a$.

But for any successive $y_n$ we need $f(s, y_{n-1})$ to be continuous, which is guaranteed if $|y_{n-1}| < b$.

Define $M$ as the maximum value of $|f|$ in $R$, which also means that $|y'| \leq M$ in $R$, i.e. $M$ is the maximum slope of $y$ in $R$.

We can use this to make sure that any $y_k(x) \leq b$. Note that $y_k(0) = 0$ for any $k$, then $|y_k(x)|$ will be less than $b$ as long as $|x| \leq h$ if we define $h$ as

$$h = \begin{cases} a & \text{if } M \leq \frac{b}{a} \\ \frac{b}{M} & \text{if } M > \frac{b}{a} \end{cases}$$

In other words, we define $|x| < a$ or $|x| < \frac{b}{M}$, whichever is smaller.

So in the interval $|x| \leq h$ the $f$ is continuous, so the integral $\int_0^x f(s, y_{n-1}(s))\,\mathrm{d}s$ can be done and $y_n$ exists and is continuous for each $n$.

The next step is to prove that the sequence converges.

Start by rewriting

$$y_n = y_0 + (y_1 - y_0) + \cdots + (y_n - y_{n-1}) = y_0 + \sum_{k=0}^{n-1} (y_{k+1} - y_k)$$

Note:

$$y_{k+1} - y_k = \int_0^x [f(s, y_k(s)) - f(s, y_{k-1}(s))] \, \mathrm{d}s$$

Since $f$ is continuous in $R$, we can write

$$|f(s, y_k(s) - f(s, y_{k-1}(s))| \leq T|(y_k(s) - y_{k-1}(s))|$$

where $T$ is the maximum value of $f_y$ in $R$ (using the mean value theorem on $f$).

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Prove by induction that $|y_n - y_{n-1}| \le M\, T^{n-1}\, \frac{|x|^n}{n!}$

For $n = 1$: $|y_1 - 0| = |\int_0^x f(s,0)\,\mathrm{d}s| \le M|x|$, since $y_0 = 0$.

Next assume that $|y_{n-1} - y_{n-2}| \le M\, T^{n-2}\, \frac{|x|^{n-1}}{(n-1)!}$

Check: $|y_n - y_{n-1}| = |\int_0^x [f(s, y_{n-1}(s)) - f(s, y_{n-2}(s))]\,\mathrm{d}s| \le$

$T \int_0^x |y_{n-1}(s) - y_{n-2}(s)|\mathrm{d}s \le T\, M\, T^{n-2} \int_0^x \frac{|s|^{n-1}}{(n-1)!}\mathrm{d}s \le M\, T^{n-1}\, \frac{|x|^n}{n!}$

and the proof by induction is complete.

We had: $y_n = y_0 + \sum_{k=0}^{n-1}(y_{k+1} - y_k)$

Taking the absolute value:

$$|y_n| \leq |y_0| + \sum_{k=0}^{n-1} |y_{k+1} - y_k| \leq |y_0| + \sum_{k=0}^{n-1} M\, T^k\, \frac{|x|^{k+1}}{(k+1)!}$$

and since $|x| \leq h$:

$$|y_n| \leq \frac{M}{T} \sum_{k=0}^{n-1} \frac{(Th)^{k+1}}{(k+1)!}$$

The last series converges uniformly, so the sequence $y_n$ also converges uniformly. (Continued on next lecture)

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